

Abhishek Sharma

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RESEARCH INTERESTS

Systems for large-scale AI/ML workloads — distributed and replicated systems (consensus, fault tolerance, linearizability), infrastructure for LLM serving and orchestration, and performance/reliability engineering for data-intensive and high-performance computing pipelines.

EDUCATION

- **IOE Pulchowk Campus, Lalitpur, Nepal** Nov. 2019 – Apr. 2024
Bachelor in Computer Engineering 76% average
- **SOSHGS, Gandaki, Pokhara, Nepal** 2017 – 2019
+2 Major in Physics and Mathematics 3.85 GPA (CGPA 3.72)

PUBLICATIONS

Topic Modeling for Nepali Political News — *IEEE ICICT 2025* | [Paper](#) | [Code](#)
Compared probabilistic (LDA), algebraic (NMF, LSA), and transformer-based (BERTopic) methods on a 2018–2023 Nepali political news corpus in a low-resource setting.

RESEARCH EXPERIENCE

- Undergraduate Researcher (B.E. Final-Year Project)** 2024
IOE Pulchowk Campus | Supervisor: *Dr. Aman Shakya*
 - Worked on a low-resource NLP project on Nepali political news using probabilistic, algebraic, and transformer-based methods. Published at IEEE ICICT 2025 (see Publications).
- Product & Technical Design** 2025 – 2026
ZeroTB
 - Translated business requirements and stakeholder input into engineering flows, system architecture, and schema design for an AI-powered compliance-automation product, informed by competitor and USP analysis.

PROJECTS

- Quasar: Linearizable Replicated Key-Value Store** | [Link](#) | *Raft, SMR, WAL, FastAPI, Python*
 - Implemented a Raft replicated state machine with term-based leader election, log-matching replication, majority-quorum commit, plus crash-durable WAL, snapshot compaction, and InstallSnapshot catch-up under crash and partition.
 - Served linearizable reads via a majority quorum check so an isolated leader cannot return stale state; built an interactive lab to step through election, partition, crash, and stale-leader recovery on a live three-node cluster.
- Videolize: Text Script to Video Generator** | [Link](#) | *Stable Diffusion, Gemini API*
 - Script-to-video pipeline using Stable Diffusion, spherical latent interpolation, and Gemini API; 1st prize, GenAI category, Locus Hack-A-Week 2023.
- Information Extraction from Invoices using LLM** | *GPT-4 Vision, YOLO, Table-Transformers*
 - Table and key-value extraction with GPT-4 Vision and fine-tuned Table-Transformers/YOLO; 1st prize, LLM category, Kathmandu University AI Competition 2023.
- Checkers** | [Link](#) | *C++, Minimax, Alpha-beta pruning*
 - C++ board-game engine with Minimax and alpha-beta pruning as an AI opponent.
- Procedural Terrain Generator** | [Link](#) | *C++, Perlin noise*
 - Hill terrain from Perlin noise, with controls for randomness, clipping, and wireframe rendering.
- University Network Infrastructure Design** | [Link](#) | *Cisco Packet Tracer*
 - Campus topology with subnetting, VLANs, DNS/web servers, and OSPF dynamic routing in Cisco Packet Tracer.

TEACHING EXPERIENCE

Instructor

2023

LOCUS Fellowship, Pulchowk Campus

- Co-designed curriculum and taught the mathematical foundations of machine learning (linear algebra, optimization, supervised algorithms) to 100+ students, including implementations from first principles.

Programming Instructor

2021 – 2022

IT Club and PDSC, Pulchowk Campus

- Taught Django web development to 250+ students (PDSC) and Git/GitHub workflows to 40+ students in an in-person IT Club workshop.

PROFESSIONAL EXPERIENCE

Founding Software Engineer

April 2025 – Present

ZeroTB

San Francisco, CA / Kathmandu

- Contributed to scaling product from zero to six production clients; owned backend architecture, APIs, and early system-design decisions.
- Owned production-critical database, third-party integration, core framework engine, trust center and notification system services —schema design, performance, backend infra, and live-client migrations.
- Built RAG and LLM-orchestration workflows (LangChain, LlamaIndex) for automated compliance document processing.
- Contributed engineering architecture, system flow, and technical slides for the investor pitch deck and YC applications.

Associate Software Engineer (ML)

April 2024 – April 2025

Docsumo

Chabahil, Kathmandu

- Performed critical performance analysis on the key-value extraction pipeline and reduced extraction latency by ~ 27% (7–8 seconds), significantly improving system throughput.
- Worked closely with the SRE team to improve reliability and scalability of distributed systems using GCP, Kubernetes, Docker, and Redis.
- Contributed to the core document-extraction pipeline for key-value and OMR extraction systems, including research, implementation, optimization, and refactoring of production APIs.
- Built and maintained scalable backend services and APIs using Flask and MongoDB, supporting high-volume document-processing workflows.

LEADERSHIP & MEMBERSHIP

President

Apr 2023 – Feb 2024

Society of Kaski Engineering Students (SOKES), Pulchowk Campus

- Led a student society of 200+ engineering students from Kaski District at Pulchowk Campus.

Secretary

Jan 2022 – May 2023

IT Club, Pulchowk Campus

- Department representative for Electronics and Computer Engineering; organized Git and design workshops and expanded outreach to students from other campuses.

AWARDS & ACHIEVEMENTS

Winner — LLM category in AI Competition organized by Kathmandu University [Dec. 2023] | [Link](#)

Winner — Generative AI category in LOCUS 2024 HackAWeek (Hackathon) [Jan. 2024]

Full Scholarship UnderGrad — **Rank 34** in IOE Examination (over 14000+ students) | [Link](#)

Rank 1 — Entrance Exam of Kathmandu University [Electronics Department rank] (over 10000+ students) | [Link](#)

Golden Jubilee Scholarship — Awarded By Embassy of India in Kathmandu, Nepal

Position Holder — National Level Math Competition (Budanilkantha Math League) | [Link](#)

Entrance Topper — High School examination out of over 3000+ students

College Topper — High School (Grade 12) Board Examination [**3.85 GPA**]

School Topper — SEE Level Examination [**3.8 GPA**]

CERTIFICATIONS

Machine Learning Specialization — Coursera & DeepLearning.AI | [Link](#)

Deep Learning Specialization — Coursera & DeepLearning.AI | [Link](#)

Annual Nepal AI School — NAAMII,Nepal | [Link](#)

AWS Academy Graduate - AWS Academy Cloud Architecting — AWS | [Link](#)

AWS Academy Graduate - AWS Academy Data Engineering — AWS | [Link](#)

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL, Dart, HTML/CSS

Frameworks: Flask, FastAPI, Django, PyTorch, Node.js, Express.js, Flutter

Knowledge: Linux/Unix, Distributed Systems (Raft, SMR, linearizability), LLM systems (transformer inference, KV cache, RAG, agents), Computer Networks, Web Dev, Computer Vision (YOLO, CNN, ResNets, U-Net, etc.), Mathematical Domain (Linear Algebra, Probability and Statistics,etc.), NLP (RNNs, LSTMs, Transformers), DSA

Developer Tools: Git, Docker, Kubernetes, Postman

Libraries: gensim, spaCy, NLTK, scikit-learn, NumPy, Matplotlib, seaborn, Pandas

Databases: SQL (PostgreSQL, MySQL, SQLite), NoSQL (MongoDB)

Cloud Services: GCP, AWS

LANGUAGE PROFICIENCY

English — IELTS 8.0 (Listening 8.5, Reading 8.5, Writing 7.0, Speaking 7.0)

Nepali — Native